Jenkins Exam

MCQs Questions

1️⃣ What is Jenkins mainly used for?

A) Server Monitoring

B) Continuous Integration and Continuous Delivery

C) Database Management

D) Container Orchestration

2️⃣ Which type of job allows you to define build steps using code in Jenkins?

A) Freestyle Project

B) Pipeline Project

C) Multi-Configuration Project

D) External Job

3️⃣ Which file is used to define a pipeline in Jenkins?

A) pipeline.yaml

B) dockerfile

C) Jenkinsfile

D) build.gradle

4️⃣ What is the purpose of a Jenkins Agent (Node)?

A) To store source code

B) To execute jobs assigned by the Jenkins controller

C) To manage plugins

D) To configure webhooks

5️⃣ Which plugin is required to connect Jenkins with GitHub?

A) Docker Plugin

B) Git Plugin

C) Kubernetes Plugin

D) Maven Plugin

6️⃣ What is the purpose of a Webhook in Jenkins CI/CD?

A) To install plugins

B) To trigger build automatically on code push

C) To secure Jenkins server

D) To restart Jenkins

7️⃣ Which command is used inside Jenkins Pipeline to execute shell commands?

A) bash

B) cmd

C) sh

D) run

8️⃣ What is the purpose of `post` block in Jenkins Pipeline?

A) Define environment variables

B) Execute steps after pipeline stages

C) Define agents

D) Install plugins

9️⃣ What is the use of `sshagent` in Jenkins Pipeline?

A) Install SSH on server

B) Store SSH keys

C) Use stored SSH credentials during execution

D) Restart SSH service

🔟 What happens if a stage fails in Jenkins Pipeline (by default)?

A) The pipeline continues to next stage

B) The pipeline stops execution

C) Jenkins restarts automatically

D) All stages are skipped but marked successful

Jenkins CI/CD Practical Test

You are hired as a **Junior DevOps Engineer** in a startup organization. The development team has already created two static websites and stored their code in GitHub repositories.

Your task is to set up CI/CD pipelines using Jenkins and automatically deploy both applications on the same target server.

You are responsible for complete automation and troubleshooting.

Application Repositories

Clone the following repositories:

Application 1: FoodHub Restaurant Website

Repository:

```
https://github.com/iamtruptimane/FoodHub-Restaurant-Website.git
```

Clone command:

```
git clone https://github.com/iamtruptimane/FoodHub-Restaurant-Website.git
```

Application 2: ShopEase Website

Repository:

```
https://github.com/iamtruptimane/ShopEase-Website.git
```

Clone command:

```
git clone https://github.com/iamtruptimane/ShopEase-Website.git
```

Task 1: Infrastructure Setup

Create two Linux servers:

Jenkins Server

- Install Java
- Install Jenkins
- Install required plugins
- Configure SSH credentials
- Verify Jenkins is accessible via browser

Target Server

- Install nginx
- Open port 80
- Ensure websites are accessible from browser

Task 2: Deploy Both Applications on the Same Server

Deploy both applications on the same target server.

Expected URLs:

```
http://<Target-Server-IP>/foodhub
http://<Target-Server-IP>/shopease
```

Expected deployment paths:

```
/var/www/html/foodhub
/var/www/html/shopease
```

Configure nginx accordingly.

Task 3: Create Jenkins Pipeline Jobs

Create two separate Jenkins Pipeline jobs:

1. FoodHub-Pipeline
2. ShopEase-Pipeline

Each pipeline should:

- Clone code from GitHub
- Copy files to target server
- Restart nginx if required
- Verify deployment

Task 4: Configure GitHub Webhooks

Configure GitHub webhooks for both repositories such that:

- Any push to GitHub automatically triggers Jenkins.
- Jenkins deploys the latest code to the target server.

Task 5: Make Required Changes

FoodHub Website

Modify:

```
Fresh Food Everyday
```

To:

```
Delicious Food Delivered Fast
```

Commit and push the changes.

Verify automatic deployment.

ShopEase Website

Modify:

```
Welcome to ShopEase
```

To:

```
Welcome to ShopEase Online Store
```

Commit and push the changes.

Verify automatic deployment.

Task 6: Troubleshooting

If deployment fails, investigate and resolve the issue.

Possible areas to check:

- Jenkins logs
- Build history
- SSH connectivity
- nginx status
- File permissions
- Security Group rules
- GitHub webhook delivery
- Jenkins credentials
- Deployment path
- Port accessibility

Bonus Task

Configure nginx so that:

```
http://<Target-Server-IP>/foodhub
```

opens the FoodHub website and

```
http://<Target-Server-IP>/shopease
```

opens the ShopEase website without using different ports.

Deliverables

Students must submit:

1. Jenkins pipeline screenshots
2. GitHub webhook configuration screenshot
3. Browser output of both applications
4. Jenkins build success screenshot
5. GitHub commit history
6. Target server deployment proof